THE LOCUS WHERE A FLAT FINITELY PRESENTED MODULE IS ETALE LOCALLY FREE IS OPEN AND STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5988-5999 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6001-6017, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Finitely presented modules flat over a morphism, the pointwise freeness criterion, and constructibility; the argument is retained.

Source reference. Original lemma lemma-flat-and-free-at-point-fibre: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5940–5954. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-flat-on-fibres-at-point: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5632–5662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite presentation. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module flat over Y. Then the set*

$$\{x \in |X| : \mathcal{F} \text{ free in an étale neighbourhood of } x\}$$

is open in $|X|$ and its formation commutes with arbitrary base change $Y' \to Y$.

Proof. Openness holds trivially. Let $Y' \to Y$ be a morphism of algebraic spaces, set $X' = Y' \times_Y X$, and let $x' \in |X'|$ be a point lying over $x \in |X|$. By Lemma lemma-flat-and-free-at-point-fibre we see that x is in our set if and only if $(\mathcal{F}_{\overline{y}})_{\overline{x}}$ is a flat $\mathcal{O}_{X_{\overline{y}},\overline{x}}$-module. Similarly, x' is in the analogue of our set for the pullback $\mathcal{F}'$ of $\mathcal{F}$ to X' if and only if $(\mathcal{F}'_{\overline{y}'})_{\overline{x}'}$ is a flat $\mathcal{O}_{X'_{\overline{y}'},\overline{x}'}$-module (with obvious notation). These two assertions are equivalent by Lemma lemma-flat-on-fibres-at-point applied to the morphism $\mathrm{id} : X \to X$ over Y. Thus the statement on base change holds. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.